Overview of Available Datasets

Table of contents

- Summary
- Question and Scope
- Data and Methods
- Results
  - Dataset Inventory and Characteristics
    - 1. PulseNet (CDC Molecular Surveillance Network)
    - 2. NORS (National Outbreak Reporting System)
    - 3. Census Population (US Census Bureau)
    - 4. Map the Meal Gap (Feeding America)
    - 5. Social Vulnerability Index (SVI; CDC/ATSDR)
    - 6. NACo (National Association of Counties)
  - Temporal Overlap and Common Years
  - Dataset Characteristics and Quality Assessment
  - Cross-Dataset Integration Strategy
- Interpretation and Limitations
  - Data Quality and Completeness Considerations
  - Temporal Limitations
  - Geographic Resolution and Joining Feasibility
  - Excluded Dataset
- Conclusion
- References

List of figures

- Figure 1: Temporal overlap across datasets. Horizontal rows show the temporal coverage of each dataset: PulseNet (1996–2023, 27 years), NORS (1998–2021, 23 years), Map the Meal Gap (2009–2020, 12 years), Census Population (2020–2023, 4 years), and SVI (2020, 2022, 2 years). A vertical red line marks 2020 as the most recent year with concurrent data across all five usable datasets. The gray line shows the cumulative count of datasets with available data per year, reaching a maximum of 5 in 2020.
- Figure 2: Dataset characteristics across five dimensions. The top-left panel shows temporal coverage (years), with PulseNet spanning 1996–2023 and SVI limited to 2020 and 2022. The top-right panel displays dataset size on a logarithmic scale, showing PulseNet's dominance (582,022 records) compared to other datasets (~3,000–4,000 records). The bottom-left panel presents data quality assessment scores (0–1 scale) across five metrics: FIPS code availability, geographic completeness, clean data structure, temporal coverage adequacy, and dataset size. The bottom-right panel categorizes geographic join reliability: SVI

and Map the Meal Gap are rated "Excellent" (have FIPS codes); PulseNet and NORS are "Fair" (require State/County name matching); Census is "Poor" (name-only, unreliable).

- Figure 3: Cross-dataset integration workflow. The diagram illustrates a three-step process: (1) Build a FIPS Master Lookup Table from datasets that already contain FIPS codes (SVI and Map the Meal Gap); (2) Apply the lookup table to datasets lacking FIPS codes (PulseNet, NORS, Census) to standardize geographic identifiers via State and County name matching; (3) Join all datasets using FIPS as the common geographic key. Color coding indicates join difficulty: green ("Has FIPS—Easy join") for SVI and Map the Meal Gap, orange ("No FIPS—Needs lookup") for PulseNet and NORS, and pink ("Name-only—Hardest") for Census.

List of tables

- Table 1: Summary of Available Datasets

Summary

This analysis describes six datasets available for research on foodborne disease and social determinants of health in the United States. Five datasets are usable, containing approximately 592,000 total records spanning 1996–2023 at the county and state geographic levels. The sixth dataset (NACo) contains no tabular data and must be excluded from analysis.

The datasets vary substantially in temporal coverage, geographic joining capability, size, and data structure. PulseNet dominates by record count (582,022 isolates) and temporal span (1996–2023). The most recent year with concurrent data across all five usable datasets is **2020**. Cross-dataset integration requires standardization of geographic identifiers, as only two datasets include Federal Information Processing (FIPS) codes; the others necessitate name-based lookups and joins.

Question and Scope

This report catalogs the available datasets and characterizes their key attributes: geographic coverage, temporal range, data structure, record counts, and critical data-quality considerations relevant to epidemiological analyses. The primary focus is preparing researchers to understand dataset capabilities, limitations, and integration strategies before beginning analysis.

Data and Methods

A systematic review of the six archived datasets was performed to extract metadata, load instructions, column definitions, temporal coverage, and geographic scope. Each dataset was inspected for file structure, join keys, data quality quirks, and completeness. Temporal overlap analysis was conducted to identify years with concurrent data across datasets. Geographic joining feasibility was assessed based on the availability and format of geographic identifiers (FIPS codes or State/County names).

Results

Dataset Inventory and Characteristics

Table 1: Summary of Available Datasets

Dataset	Records	Years	Geographic Level	FIPS Codes	Key Variable(s)	Status
PulseNet	582,022	1996–2023	County/State	No	Serotype, Isolate Date, Source Type	Usable
NORS	3,950	1998–2021	County/State	No	Outbreak ID, Illness Count, Hospitalizatio n, Deaths	Usable
Census Population	3,151	2020–2023	County	No	Population by year	Usable
Map the Meal Gap	~3,143/year	2009–2020	County	Yes	Food Insecurity Rate	Usable
SVI	3,143	2020, 2022	County	Yes (int)	Vulnerability Scores, 158 columns	Usable
NACo	0	Unknown	County	Unknown	N/A	**Unusable**

1. PulseNet (CDC Molecular Surveillance Network)

Data scale and scope: PulseNet contains 582,022 individual laboratory isolate records spanning 27 years (1996–2023), representing the largest dataset by record volume. Each record documents a Salmonella isolate submitted for molecular subtyping, including its serotype, collection date, source (state and county), and clinical context.

Key columns: Serotype, SourceState, SourceCounty, IsolatDate, Outbreak (indicator of association with known outbreak), SourceType (human, animal, food, or environmental), PatientSex, and PatientAge.

Geographic coverage: County and state level; however, county information is missing for a substantial portion of records. Geographic joining requires standardization of state and county names, as FIPS codes are not provided.

Critical data-quality considerations:

- Data are split across six separate Excel files that must be concatenated for complete analysis.
- IsolatDate is stored as object dtype and requires conversion to datetime.
- SourceType and PatientSex values have inconsistent capitalization and whitespace, requiring normalization with .str.upper().str.strip().
- Large dataset size (582,022 records) may create memory constraints in some analyses.
- Mix of clinical (human) and non-clinical (food, animal, environmental) isolates; filtering by source type is often necessary.

2. NORS (National Outbreak Reporting System)

Data scale and scope: NORS contains 3,950 documented foodborne disease outbreak events from 1998–2021 (23 years), collected and maintained by the CDC. This is a complex relational database comprising 39 interconnected Excel sheets linked by a unique CDCID (outbreak identifier).

Key columns (NORSMain sheet): CDCID (unique outbreak ID), yearfirstill, monthfirstill, ExposureState, ExposureCounty, IllCount (number of illnesses), HospCount (hospitalizations), and DeathCount (deaths).

Supplementary sheets: GenEtiology (serotype information), Foods (implicated food vehicles), Settings (exposure locations), and Water (water-related exposures). Serotype data are not in the main sheet and must be joined via CDCID.

Geographic coverage: State and county level; however, ExposureCounty contains '***' for suppressed counties (privacy protection), reducing geographic resolution for privacy-sensitive outbreaks.

Critical data-quality considerations:

- Complex relational structure requiring joins across 39 sheets via CDCID key.
- Serotype information requires joining GenEtiology sheet to NORSMain sheet; serotypes are not in the main table.
- County suppression ('***') reduces geographic granularity and must be handled in analysis (typically filtered out).
- Dataset ends at 2021 and does not include 2022–2023 data.
- Not all outbreaks have complete information across all 39 sheets; data are sparse in some sheets.
- Geographic joining requires standardization of state and county name strings; FIPS codes are not provided.

3. Census Population (US Census Bureau)

Data scale and scope: The Census dataset contains annual population estimates for 3,151 US counties, covering 2020–2023 (4 years). Data are from the official US Census Bureau's annual county population estimates.

Key columns: Unnamed: 0 (county and state name concatenated), 2020, 2021, 2022, 2023 (population columns). Column headers are non-standard; the county name is in the first column without a header.

Geographic coverage: County level; no FIPS codes are provided. County names have a leading period (.) that must be stripped for reliable joining with other datasets.

Critical data-quality considerations:

- **No FIPS codes available**—the most significant limitation for joining. Geographic linking relies on county name matching, which is error-prone due to naming variations (e.g., "St. Louis" vs. "Saint Louis").
- County names include a leading period (e.g., .Cook County, IL) that must be stripped with .str.lstrip('.').
- Only 4 years of data (2020–2023), providing no historical population denominators before 2020.
- Loading requires specifying header=3 in pd.read_excel() due to multiple header rows in the Excel file.
- Name-based geographic joins are inherently less reliable than FIPS-based joins and have higher error rates.

4. Map the Meal Gap (Feeding America)

Data scale and scope: Feeding America's Map the Meal Gap contains county-level food insecurity rates and related economic measures for approximately 3,143 US counties annually, covering 2009–2020 (12 years). The data are released with a 2-year publishing lag (e.g., the 2011 publication reports 2009 data).

Key columns (vary by year): FIPS (5-digit county code, consistent across years), State, County, and food insecurity rates. Column names for insecurity rates change annually (e.g., "Overall Food Insecurity Rate," "Food Insecurity Rate," "% Food Insecure"); consistent identification requires dynamic searching for columns containing "insecurity."

Geographic coverage: County level via FIPS code. This dataset is one of only two with reliable FIPS codes, making it ideal for joins.

Critical data-quality considerations:

- **2-year publication lag:** File naming follows the pattern MMG_{publication_year}_{data_year}Data_ToShare.xlsx. For example, MMG_2011_2009Data_ToShare.xlsx contains 2009 data published in 2011. This lag must be carefully tracked in temporal analyses.
- Column names and sheet names **change every year**, requiring year-specific loading and dynamic column identification. Static code will not work across all years.
- The 2018 file is an exception: it requires header=1 instead of the default header=0.
- The 2022 file is a special case: it combines both 2019 and 2020 data in a single file, resulting in 6,285 rows instead of the typical ~3,143.
- Inconsistent data structure across years necessitates careful inspection and year-specific handling code for reliable integration.
- FIPS codes are stored as strings with zero-padding and should be preserved in that format for joins.

5. Social Vulnerability Index (SVI; CDC/ATSDR)

Data scale and scope: The SVI provides comprehensive county-level vulnerability metrics for 3,143 US counties. Data are available for two years only: 2020 and 2022. The SVI ranks counties on 15 social factors organized into four themes: Socioeconomic Status, Household Composition & Disability, Minority Status & Language, and Housing Type & Transportation.

Key columns: FIPS (5-digit county code, stored as integer), STATE, ST_ABBR, COUNTY, E_TOTPOP (total population estimate), and 158 columns representing vulnerability measures. Key summary measures include RPL_THEMES (overall vulnerability ranking: 0 = lowest vulnerability, 1 = highest) and four theme-specific rankings: RPL_THEME1 (socioeconomic), RPL_THEME2 (household composition), RPL_THEME3 (minority status & language), and RPL_THEME4 (housing & transportation).

Column naming convention: SVI uses a systematic prefix system across 158 columns:

- **E_** = Estimate (raw count)
- **EP_** = Percentage
- **EPL_** = Percentile rank (0–1 scale)
- **RPL_** = Overall ranking (0–1 scale; 0 = least vulnerable, 1 = most vulnerable)

Geographic coverage: County level via FIPS code, making SVI ideal for joins with datasets containing FIPS codes.

Critical data-quality considerations:

- FIPS is stored as integer type and **must be zero-padded to 5 digits** for reliable joining: .astype(str).str.zfill(5).
- Missing values are coded as **–999** and must be filtered before analysis.

- **Only two years of data (2020, 2022)** severely limit temporal trend analysis. No historical data are available.
- RPL values represent percentile rankings where 0 = least vulnerable and 1 = most vulnerable. This counterintuitive scale (higher values = worse outcomes) can be easily misinterpreted.
- Based on American Community Survey 5-year estimates, which have inherent smoothing of annual variation.
- 158 columns require careful documentation and strategic selection for analysis; column prefix understanding is critical for correct interpretation.

6. NACo (National Association of Counties)

Status: This dataset contains no tabular data and must be excluded from all analyses. The repository contains only a links.txt file with external URLs. Do not attempt to load or process this dataset.

Temporal Overlap and Common Years

Figure 1 displays the temporal coverage of each dataset and shows the number of datasets with concurrent data in each year.

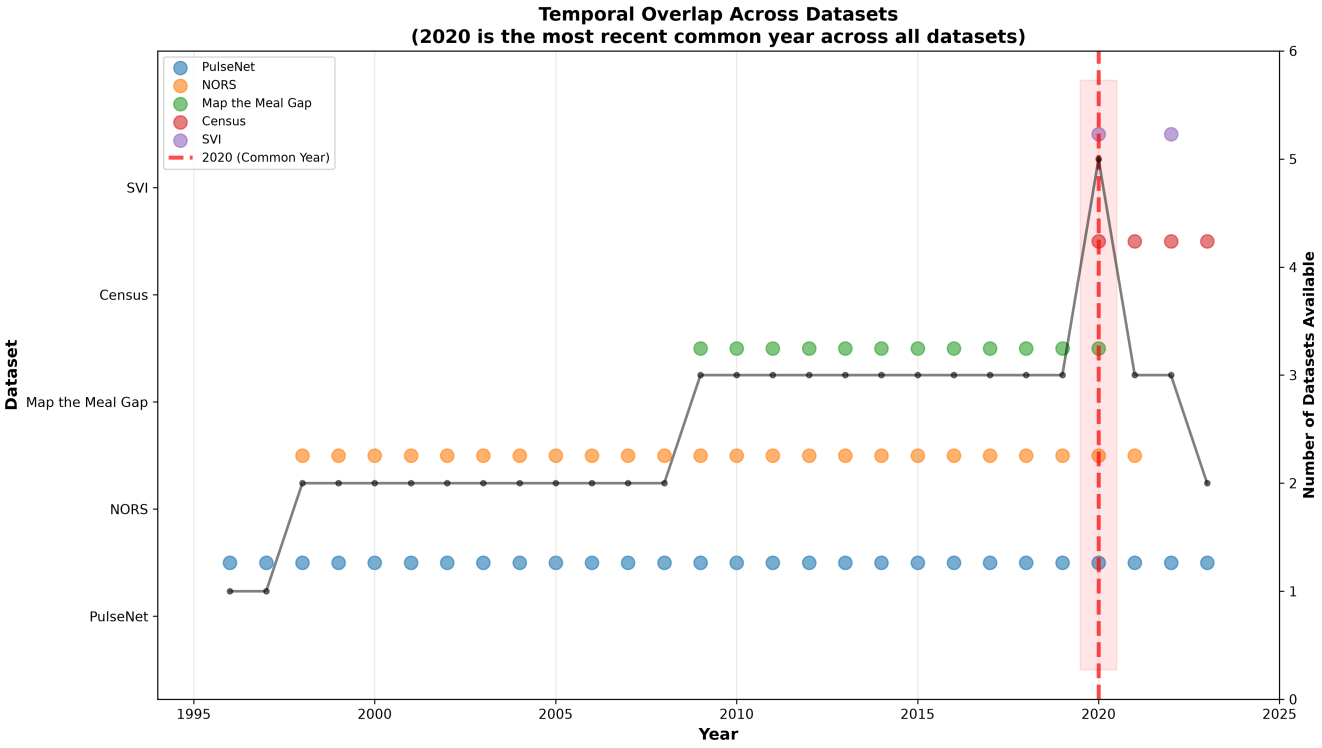


Figure 1: Temporal overlap across datasets. Horizontal rows show the temporal coverage of each dataset: PulseNet (1996–2023, 27 years), NORS (1998–2021, 23 years), Map the Meal Gap (2009–2020, 12 years), Census Population (2020–2023, 4 years), and SVI (2020, 2022, 2 years). A vertical red line marks 2020 as the most recent year with concurrent data across all five usable datasets. The gray line shows the cumulative count of datasets with available data per year, reaching a maximum of 5 in 2020.

Key finding: The year **2020 is the most recent common year across all five usable datasets**, representing the maximum temporal overlap. Before 2020, the maximum overlap was three datasets (PulseNet, NORS, and Map the Meal Gap, spanning 2009–2020). After 2020, overlap decreases: PulseNet and Census extend to 2021–2023, but NORS ends at 2021 and Map the Meal Gap ends at 2020. SVI includes only 2022 data after 2020.

Temporal coverage by dataset:

- **PulseNet:** 1996–2023 (longest coverage: 27 years)
- **NORS:** 1998–2021 (23 years)
- **Map the Meal Gap:** 2009–2020 (12 years)
- **Census Population:** 2020–2023 (4 years only)
- **SVI:** 2020 and 2022 only (2 snapshot years)

Dataset Characteristics and Quality Assessment

Figure 2 presents a multidimensional comparison of dataset characteristics across five dimensions: temporal coverage, dataset size, data quality metrics (including clean structure and geographic completeness), and geographic join capability.

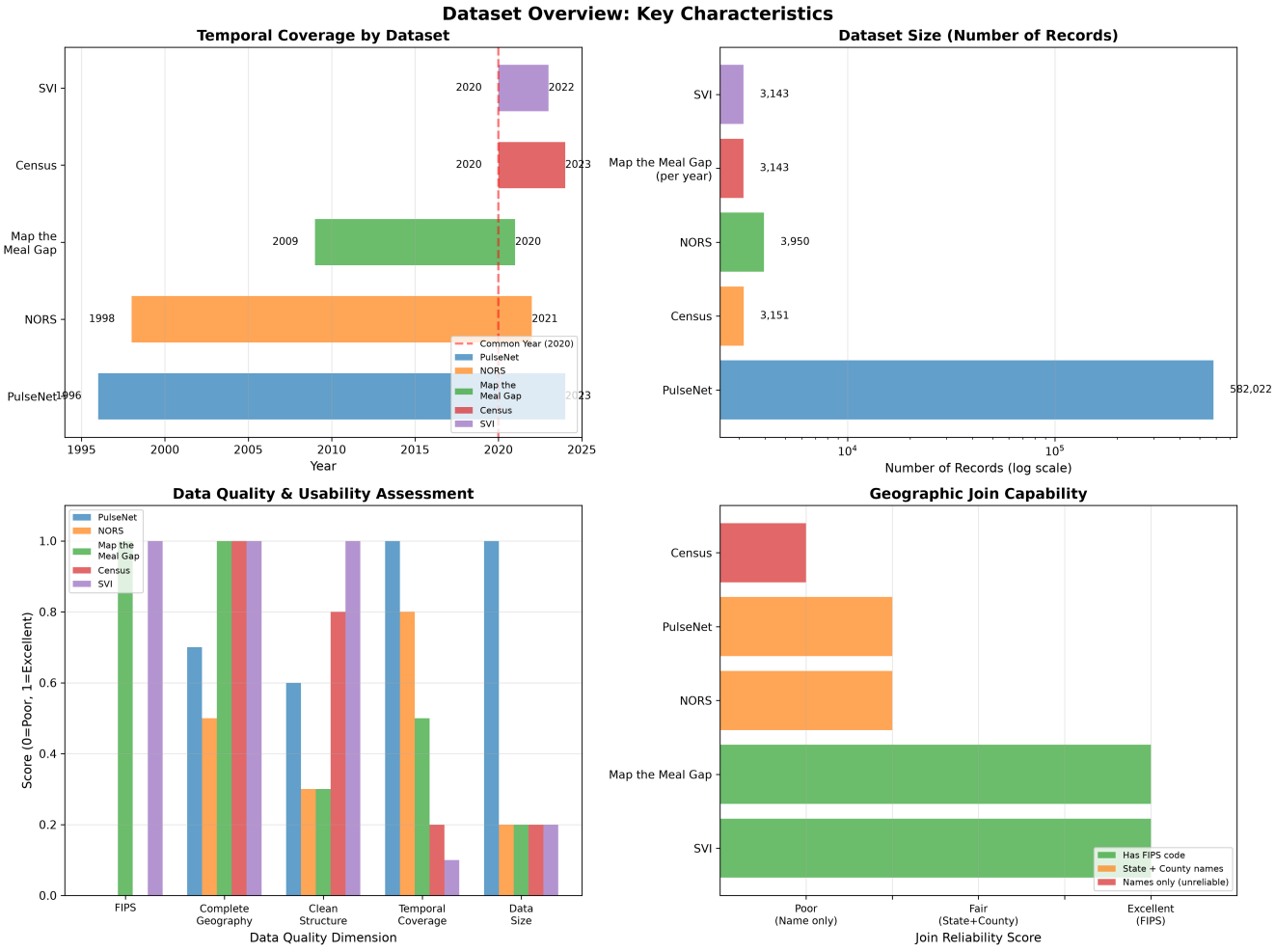


Figure 2: Dataset characteristics across five dimensions. The top-left panel shows temporal coverage (years), with PulseNet spanning 1996–2023 and SVI limited to 2020 and 2022. The top-right panel displays dataset size on a logarithmic scale, showing PulseNet's dominance (582,022 records) compared to other datasets (~3,000–4,000 records). The bottom-left panel presents data quality assessment scores (0–1 scale) across five metrics: FIPS code availability, geographic completeness, clean data structure, temporal coverage adequacy, and dataset size. The bottom-right panel categorizes geographic join reliability: SVI and Map the Meal Gap are rated "Excellent" (have FIPS codes); PulseNet and NORS are "Fair" (require State/County name matching); Census is "Poor" (name-only, unreliable).

Dataset size: PulseNet contains vastly more records (582,022) than the other datasets, which each contain approximately 3,000–4,000 records. This ~150-fold difference reflects PulseNet's isolate-level granularity

versus the county-year-level aggregation in other datasets.

Geographic join capability: Datasets with FIPS codes (SVI, Map the Meal Gap) enable straightforward geographic joins and are rated "Excellent." Datasets with only State and County name fields (PulseNet, NORS) require name-based lookup tables and are rated "Fair" due to higher error risk. Census lacks FIPS codes and relies on name matching only, earning a "Poor" rating for join reliability.

Cross-Dataset Integration Strategy

Figure 3 outlines the recommended workflow for integrating datasets with varying FIPS code availability.

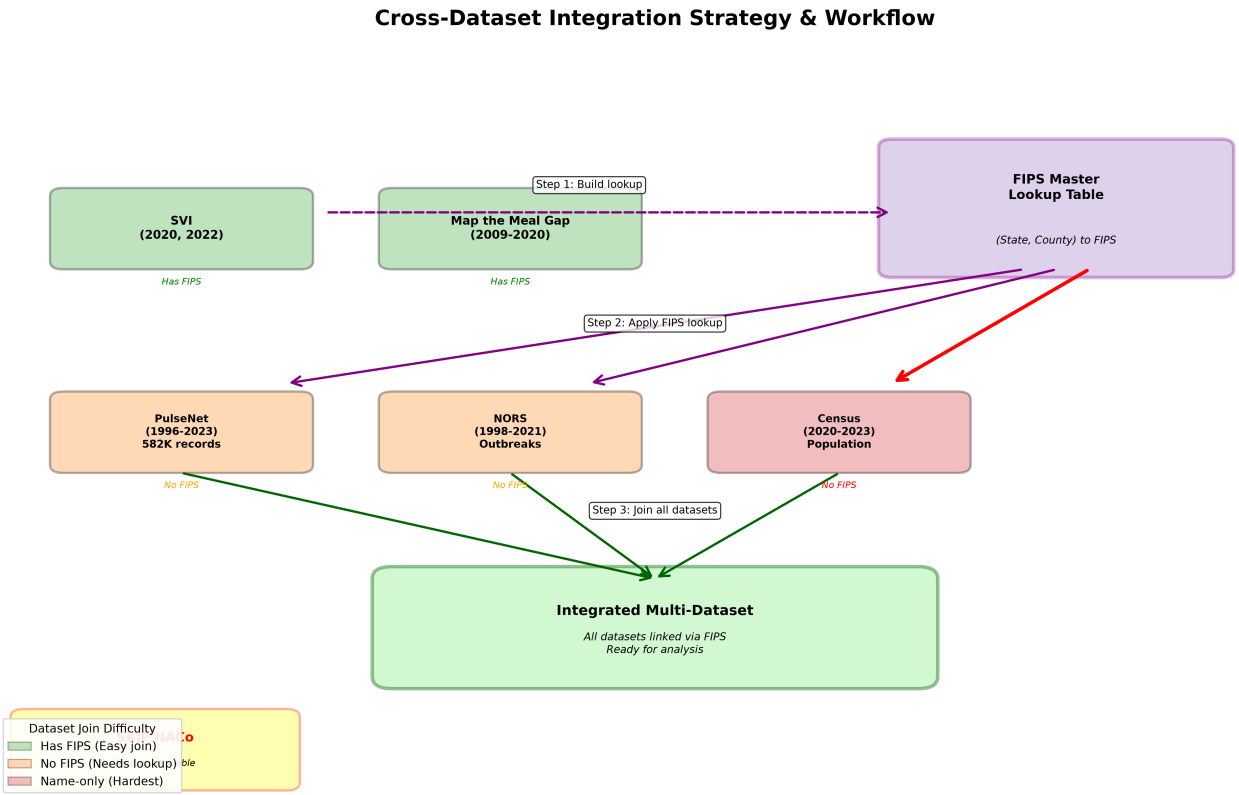


Figure 3: Cross-dataset integration workflow. The diagram illustrates a three-step process: (1) Build a FIPS Master Lookup Table from datasets that already contain FIPS codes (SVI and Map the Meal Gap); (2) Apply the lookup table to datasets lacking FIPS codes (PulseNet, NORS, Census) to standardize geographic identifiers via State and County name matching; (3) Join all datasets using FIPS as the common geographic key. Color coding indicates join difficulty: green ("Has FIPS—Easy join") for SVI and Map the Meal Gap, orange ("No FIPS—Needs lookup") for PulseNet and NORS, and pink ("Name-only—Hardest") for Census.

Integration challenges and recommendations:

1. **FIPS code standardization:** SVI stores FIPS as integers (must zero-pad to 5 digits), while Map the Meal Gap provides FIPS as zero-padded strings. Both formats are compatible after standardization.
2. **Geographic name matching:** PulseNet and NORS require joining via State and County names matched to the FIPS lookup table created from SVI and Map the Meal Gap. This introduces matching error; validation should target >95% successful match rate.
3. **Census geography:** Census county names include a leading period and require stripping (.str.lstrip('.')) and standardization before name-based joining. Name variations (e.g., "St." vs. "Saint") are common and necessitate fuzzy matching or manual review for unmatched records.

4. **Recommended join order:**
    - Begin with SVI and Map the Meal Gap (both have FIPS; direct join is reliable).
    - Create a (State, County) FIPS lookup table from these datasets.
    - Apply the lookup to PulseNet and NORS using standardized (State, County) names.
    - Join Census using the same lookup; validate Census match rate due to name variability.
    - Verify all joins; document and review unmatched records.

Interpretation and Limitations

Data Quality and Completeness Considerations

PulseNet: The dataset is complete as provided but requires preprocessing: data must be concatenated across six files, dates must be converted to datetime format, and text fields must be normalized. Large size may require memory optimization or chunking for some operations. Missing county information for some records reduces geographic resolution.

NORS: The complex 39-sheet relational structure requires careful joins via CDCID. Serotype information resides in a separate sheet and must be joined deliberately. County suppression (***) reduces geographic precision and typically results in record exclusion for county-level analyses. Dataset does not include 2022–2023 data.

Census Population: The most significant limitation is the absence of FIPS codes. Name-based joining is error-prone and typically yields <95% match rates. County name variations and leading periods require careful cleaning. Only 4 years of data (2020–2023) prevent historical population denominator assignment pre-2020.

Map the Meal Gap: Inconsistent column names and sheet structure across 12 annual files require year-specific loading logic. The 2-year publication lag must be carefully tracked. The 2022 file's anomalous structure (combining 2019 and 2020) requires special handling. Despite these quirks, FIPS code availability simplifies geographic joining.

SVI: The most severe limitation is availability of only two years (2020, 2022), which precludes temporal trend analysis within SVI. The −999 missing value code requires filtering. FIPS zero-padding requirement is easily overlooked, causing join failures. The 158-column structure, while comprehensive, demands careful column selection and documentation.

Temporal Limitations

No single dataset provides complete historical coverage. PulseNet offers the longest time series (1996–2023, 27 years) but lacks FIPS codes. Census and SVI cover only 2020–2023 and 2020–2022, respectively, providing no historical context. Researchers cannot reliably compute population-standardized rates before 2020 across all datasets. The year 2020 represents the only recent year with concurrent data across all five usable datasets.

Geographic Resolution and Joining Feasibility

Half of the datasets (PulseNet, NORS, Census) lack FIPS codes, requiring geographic joining via standardized State and County name matching. This approach is inherently error-prone: county name variations (abbreviations, spelling differences, jurisdictional changes over time) result in unmatched records. Fuzzy matching algorithms may improve match rates but require careful validation. County suppression in NORS further limits geographic precision for certain outbreaks.

Excluded Dataset

NACo contains no tabular data and cannot be used in analysis.

Conclusion

Six datasets are available for research on foodborne disease surveillance and social determinants of health. Five datasets are usable, containing ~592,000 records spanning 1996–2023 at the county and state level. PulseNet (582,022 isolates, 1996–2023) is the largest and longest-running dataset, ideal for temporal surveillance trends and serotype distribution analysis. NORS (3,950 outbreaks, 1998–2021) provides detailed outbreak-level epidemiology but ends at 2021 and requires complex joins across 39 relational sheets.

Census Population (3,151 counties, 2020–2023), Map the Meal Gap (3,143 counties/year, 2009–2020), and SVI (3,143 counties, 2020–2022) provide complementary social, economic, and demographic context. Census and Map the Meal Gap enable population-based rate calculation and food insecurity analysis. SVI supplies comprehensive vulnerability indices across four themes and 158 derived measures.

The year 2020 is the most recent year with concurrent data across all five usable datasets, limiting integrated analysis to this single year or to subsets of datasets available in other years.

Geographic joining is the primary integration challenge: only Map the Meal Gap and SVI include reliable FIPS codes. PulseNet, NORS, and Census require name-based lookups via standardized State and County fields, with elevated error risk. A three-step workflow—build a FIPS master lookup from datasets with FIPS codes, apply it to datasets lacking FIPS codes, and join all datasets via FIPS—is recommended for reliable integration.

Before beginning analysis, researchers must account for dataset-specific data-quality quirks: PulseNet requires file concatenation and text normalization; NORS requires joining across 39 sheets and filtering suppressed counties; Census requires county name cleaning and lacks FIPS codes; Map the Meal Gap has inconsistent annual structure and 2-year publication lag; SVI requires FIPS zero-padding and filtering of −999 missing values.

The NACo dataset contains no usable data and must be excluded from all analyses.

References

[1] CDC. PulseNet: The National Molecular Subtyping Network for Foodborne Disease Surveillance. Centers for Disease Control and Prevention.

[2] CDC. National Outbreak Reporting System (NORS). Centers for Disease Control and Prevention.

[3] U.S. Census Bureau. County Population Estimates. U.S. Department of Commerce.

[4] Feeding America. Map the Meal Gap. Feeding America.

[5] CDC/ATSDR. Social Vulnerability Index (SVI). Agency for Toxic Substances and Disease Registry.